
Analysis on Social Media Usage in Teens

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We live in a time where smartphones and social media are deeply integrated into daily life, especially for teenagers. Today's teenagers have unparalleled access to digital platforms, creating constant connectivity that can be both engaging and potentially harmful.

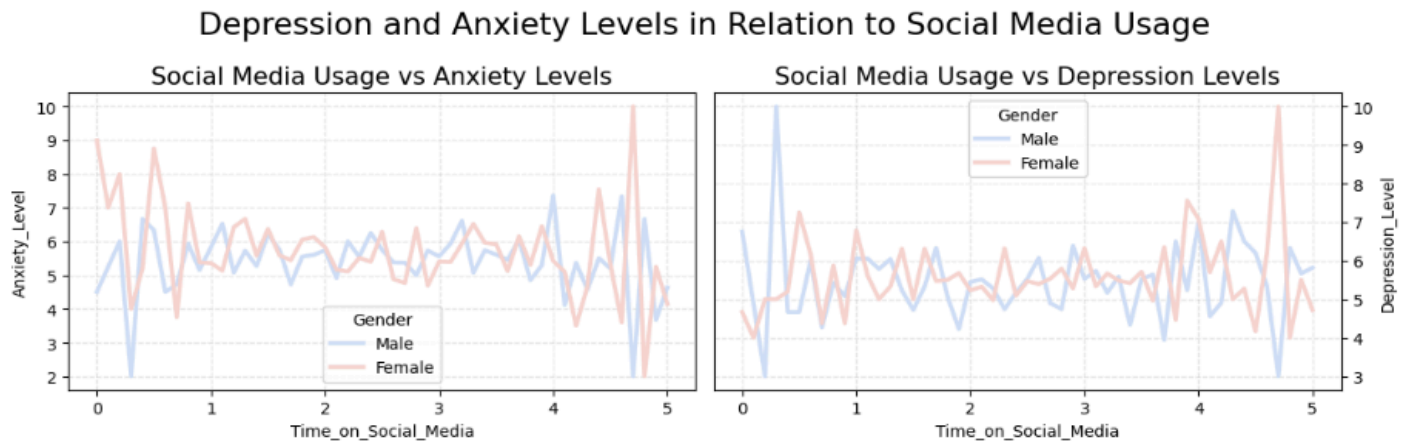
Thus, this project examines how teens aged 13–19 use their phones and investigates how these behaviors relate to mental health and indicators of addiction. To better understand these dynamics, this analysis focuses on three main questions:

1. Is there a relationship between excessive phone use and mental health deterioration?
2. How does social media usage influence factors related to mental health and addiction in teenagers?
3. Are there gender-based differences in how teenage boys and girls experience or respond to social media usage?

The goal of this analysis is to visualize these relationships and interpret the patterns that emerge from the visuals.

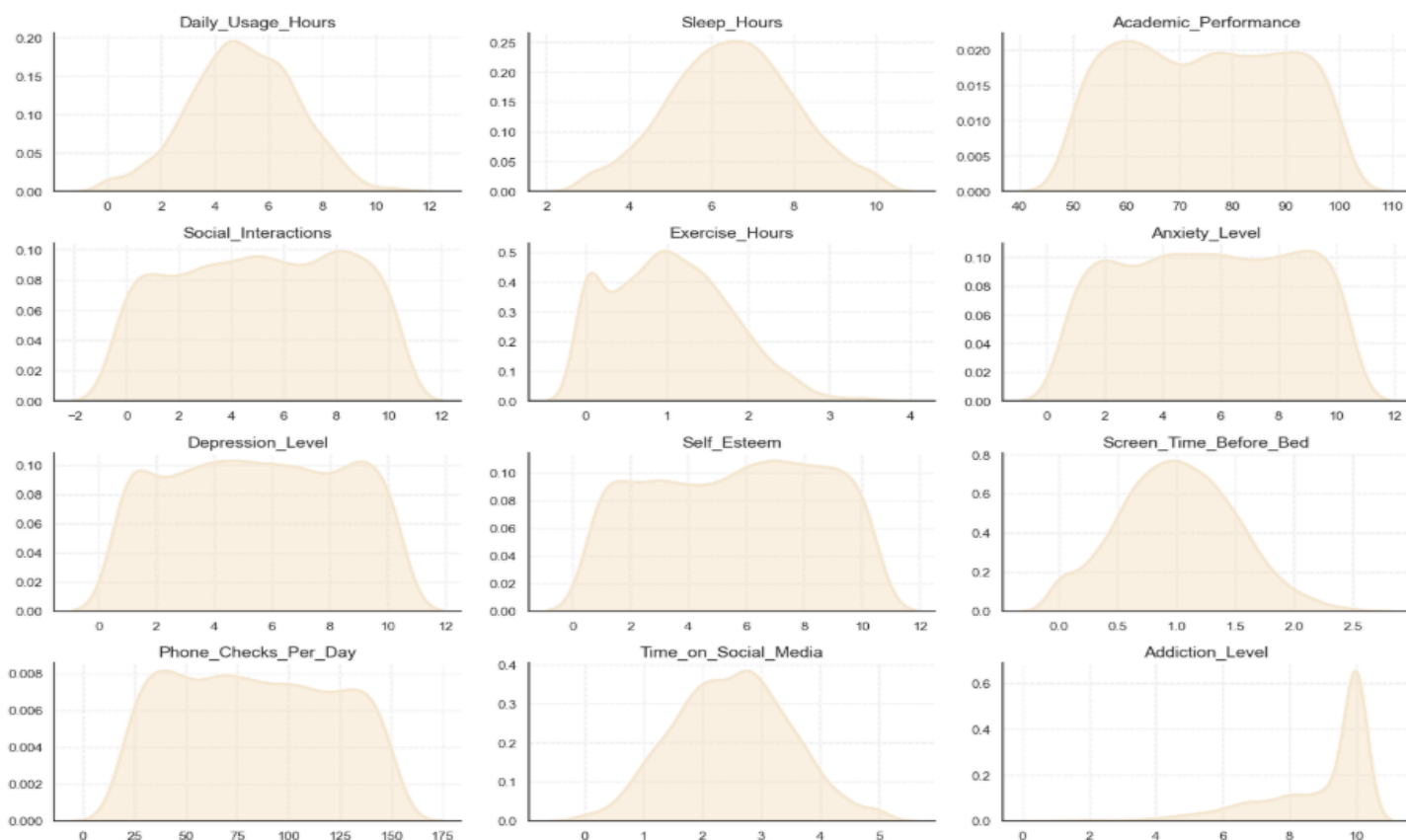
Relationship Between Phone Use and Mental Health

The first part of the analysis explores whether increased phone or social media use coincides with higher levels of anxiety or depression. While the dataset does not produce a strong or consistent correlation overall, several noteworthy patterns emerged.



Female teens display noticeable spikes in anxiety and depression between 4.5 and 5 hours of daily social media use. These spikes are far less pronounced among male teens, suggesting that heavy usage may have a different emotional impact on girls. Although these findings are not conclusive, they highlight potential gender-specific vulnerabilities worth further investigation.

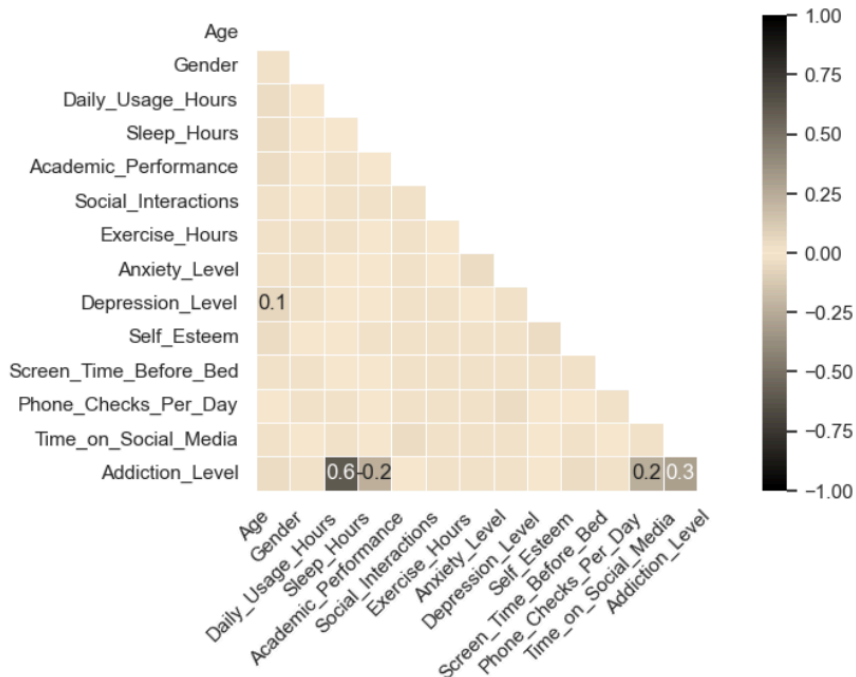
Aside from these isolated spikes, anxiety and depression levels remained relatively stable across most usage levels. Overall, mental health scores tend to be high regardless of screen time, prompting a broader review of related behavioral variables such as sleep hours, academic performance, social interactions, exercise, self-esteem, and daily phone checks.



How Social Media Usage Relates to Addiction

A key finding from the descriptive analysis is that teen addiction levels are exceptionally high. The distribution of addiction scores heavily concentrates between 8 and 10 out of 10, indicating a strong reliance on phones across the teenage group. Several notable relationships emerged:

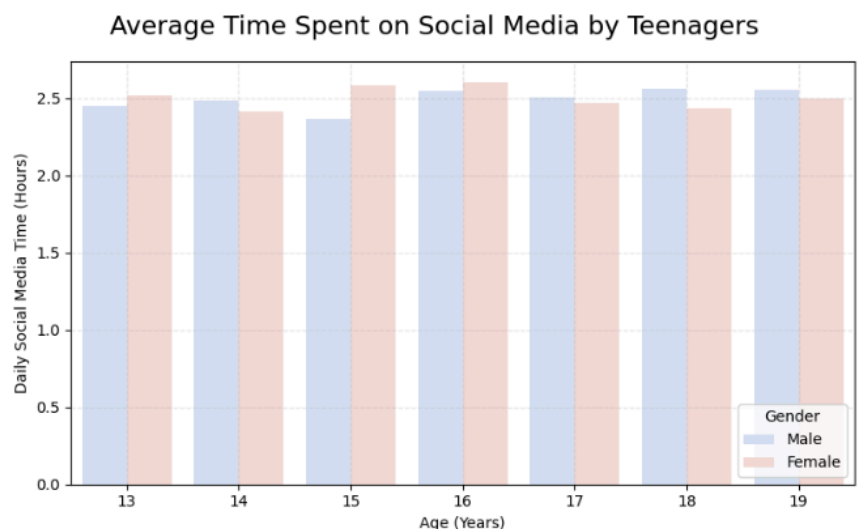
- Addiction level correlates positively with daily phone usage.
- Addiction level correlates negatively with sleep hours.
- Time spent on social media and the number of daily phone checks both correlate with higher addiction levels.



These relationships suggest that social media usage is directly tied to elevated phone addiction. Addiction levels consistently rise as social media usage increases. With a correlation of approximately 30%, this relationship is significant and provides evidence that social media contributes to addictive digital habits among teens.

Gender-Based Differences in Social Media Usage

Lastly, this analysis explores whether male and female teens differ in the amount of time they spend on social media. However, the result is no substantial gender-based differences. This suggests that high engagement with social media is widespread across teens, regardless of gender, and may reflect generational rather than gender-specific patterns.



Conclusion

This analysis highlights that smartphones and social media play a dominant role in teenagers' lives, with consistently high addiction scores and frequent phone-checking behaviors across the sample. Although the data does not reveal a direct or strong link between phone usage and mental health deterioration, the elevated addiction levels themselves raise concerns.

These findings emphasize the importance of looking beyond simple screen time metrics. Understanding the context, frequency, and nature of teens' digital behaviors is essential for identifying when healthy engagement may be shifting into problematic use. The visual analytics approach used in this project offers a clear view of these underlying patterns and can help inform future discussions among parents, educators, and policymakers about fostering balanced and supportive digital habits.

References:

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